

Solo Project 1 Climate

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Goal of Project:

In this project I will examine air quality patterns for California counties between 2018 and 2024 to determine the impact COVID-19 lockdowns had on ozone pollution. Aims to determine the places where the greatest levels of ozone pollution exist and what the health consequences are of experiencing poor air quality for an extended period. By examining location and time patterns, the research will determine which places tend to experience the poorest air quality and quantify the consequence on citizens through the use of indicators such as unhealthy air days and instances of exceeding the AQI. By merging maps indicating the dispersal of pollution with time trend analysis to provide the full scope of California's air quality challenges around this significant period that included shifts in behaviors due to the pandemic.

```
# install.packages("tidyverse")
# install.packages("lubridate")
# install.packages("RSQLite")
# install.packages("dplyr")
# install.packages("ggplot2")
# install.packages("lubridate")
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```

library(lubridate)
library(RSQLite)
library(dplyr)
library(ggplot2)
library(lubridate)

all_years_data <- bind_rows(
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
  read.csv("C:/Users/csola/OneDrive/Desktop/Pstat Solo Project 1/2025_2018_annual_aqi_by_coun
)

# head(all_years_data)
# colnames(all_years_data)
# print(all_years_data$State.Name == "California")
Cali_all_years_data <- filter(all_years_data, all_years_data$State == "California")
# the output was NA because there were no values that matched the condition
head(Cali_all_years_data)

```

	State	County	Year	Days.with.AQI	Good.Days	Moderate.Days
1	California	Alameda	2018	365	77	271
2	California	Amador	2018	364	288	70
3	California	Butte	2018	365	97	228
4	California	Calaveras	2018	364	52	292
5	California	Colusa	2018	365	206	131
6	California	Contra Costa	2018	365	111	236
	Unhealthy.for.Sensitive.Groups.Days			Unhealthy.Days	Very.Unhealthy.Days	
1	5			9	3	
2	6			0	0	
3	22			10	3	
4	14			6	0	
5	10			17	1	
6	7			8	3	
	Hazardous.Days	Max.AQI	X90th.Percentile.AQI	Median.AQI	Days.CO	Days.NO2
1	0	248	79	57	0	1
2	0	129	71	40	0	0
3	5	682	104	59	0	0
4	0	160	87	58	0	0

5	0	299	84	46	0	0
6	0	270	74	55	0	0
	Days.Ozone	Days.PM2.5	Days.PM10			
1	22	342	0			
2	364	0	0			
3	148	216	1			
4	71	293	0			
5	138	210	17			
6	45	320	0			

Now lets select specific columns and lets clean data.

```
# we do not include "State" column because we know the only value it has is "California"
Cali_all_years_data <- select(Cali_all_years_data, ~State)

head(Cali_all_years_data, 10)
```

	County	Year	Days.with.AQI	Good.Days	Moderate.Days
1	Alameda	2018	365	77	271
2	Amador	2018	364	288	70
3	Butte	2018	365	97	228
4	Calaveras	2018	364	52	292
5	Colusa	2018	365	206	131
6	Contra Costa	2018	365	111	236
7	Del Norte	2018	180	134	46
8	El Dorado	2018	365	221	107
9	Fresno	2018	365	61	206
10	Glenn	2018	363	242	105

	Unhealthy.for.Sensitive.Groups.Days	Unhealthy.Days	Very.Unhealthy.Days
1	5	9	3
2	6	0	0
3	22	10	3
4	14	6	0
5	10	17	1
6	7	8	3
7	0	0	0
8	27	9	1
9	79	19	0
10	9	6	1

	Hazardous.Days	Max.AQI	X90th.Percentile.AQI	Median.AQI	Days.CO	Days.NO2
1	0	248	79	57	0	1
2	0	129	71	40	0	0

3	5	682	104	59	0	0
4	0	160	87	58	0	0
5	0	299	84	46	0	0
6	0	270	74	55	0	0
7	0	80	58	33	0	0
8	0	203	101	45	0	0
9	0	185	133	75	0	0
10	0	255	71	39	0	0

	Days.Ozone	Days.PM2.5	Days.PM10
1	22	342	0
2	364	0	0
3	148	216	1
4	71	293	0
5	138	210	17
6	45	320	0
7	0	180	0
8	347	9	9
9	210	151	4
10	130	185	48

```
colnames(Cali_all_years_data)
```

[1] "County"	"Year"
[3] "Days.with.AQI"	"Good.Days"
[5] "Moderate.Days"	"Unhealthy.for.Sensitive.Groups.Days"
[7] "Unhealthy.Days"	"Very.Unhealthy.Days"
[9] "Hazardous.Days"	"Max.AQI"
[11] "X90th.Percentile.AQI"	"Median.AQI"
[13] "Days.CO"	"Days.NO2"
[15] "Days.Ozone"	"Days.PM2.5"
[17] "Days.PM10"	

```
# Days.with.AQI - Total number of days air quality was measured
#
# Good.Days - Days when AQI was 0-50 (Good air quality)
#
# Moderate.Days - Days when AQI was 51-100 (Acceptable)
#
# Unhealthy.for.Sensitive.Groups.Days - Days when AQI was 101-150 (Unhealthy for
# children, elderly, people with lung disease)
#
# Unhealthy.Days - Days when AQI was 151-200 (Everyone may experience effects)
```

```
#
# Very.Unhealthy.Days - Days when AQI was 201-300 (Health alert)
#
# Hazardous.Days - Days when AQI was 301-500 (Health warnings of emergency conditions)

# Days.CO - Number of days Carbon Monoxide was the main pollutant
#
# Days.NO2 - Number of days Nitrogen Dioxide was the main pollutant
#
# Days.Ozone - Number of days ground level Ozone was the main pollutant (The focus!)
#
# Days.PM2.5 - Number of days fine particulate matter (2.5 micrometers) was main pollutant
#
# Days.PM10 - Number of days coarse particulate matter (10 micrometers) was main pollutant
```

understand data more

```
# AQI is the air quality index, the higher the number, the worse the air quality
summary(Cali_all_years_data$Median.AQI)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
11.00	41.00	47.50	49.59	55.00	93.00

```
summary(Cali_all_years_data$Max.AQI)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
38.0	115.0	163.0	270.7	228.5	8368.0

```
#X90th.Percentile.AQI is The AQI value that is higher than 90% of the daily AQI readings for
summary(Cali_all_years_data$X90th.Percentile.AQI)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
31.00	61.00	77.00	83.66	97.00	224.00

```
summary(Cali_all_years_data$Days.Ozone)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	133.8	185.0	180.4	226.5	365.0

```
range(Cali_all_years_data$Year)
```

```
[1] 2018 2024
```

```
# > 2018-2024
```

```
sum(is.na(Cali_all_years_data))
```

```
[1] 0
```

```
# there is no NA values in the dataset
```

Now lets see which California countries have the worst ozone levels.

```
# n() is a function that counts the number of rows in each group
# we ungroup near the end because we want to make sure that the data is not grouped anymore
```

```
cali_county_ozone <- Cali_all_years_data |>
  group_by(County) |>
  summarise(
    avg_ozone_days = as.integer(mean(Days.Ozone)),
    total_ozone_days = sum(Days.Ozone),
    max_ozone_aqi = max(Max.AQI),
    avg_unhealthy_days = as.integer(mean(Unhealthy.Days)),
    avg_sensitive_group_days = as.integer(mean(Unhealthy.for.Sensitive.Groups.Days)),
    avg_very_unhealthy_days = as.integer(mean(Very.Unhealthy.Days)),
    n_years_of_data = n() # Number of years with data
  ) |>
  ungroup() |>
  arrange(desc(avg_ozone_days))

print(cali_county_ozone)
```

```
# A tibble: 53 x 8
```

	County	avg_ozone_days	total_ozone_days	max_ozone_aqi	avg_unhealthy_days
	<chr>	<int>	<int>	<int>	<int>
1	Amador	360	2526	156	0
2	El Dorado	350	2456	375	3
3	Lake	344	2413	233	0
4	Tuolumne	330	2313	154	0

```

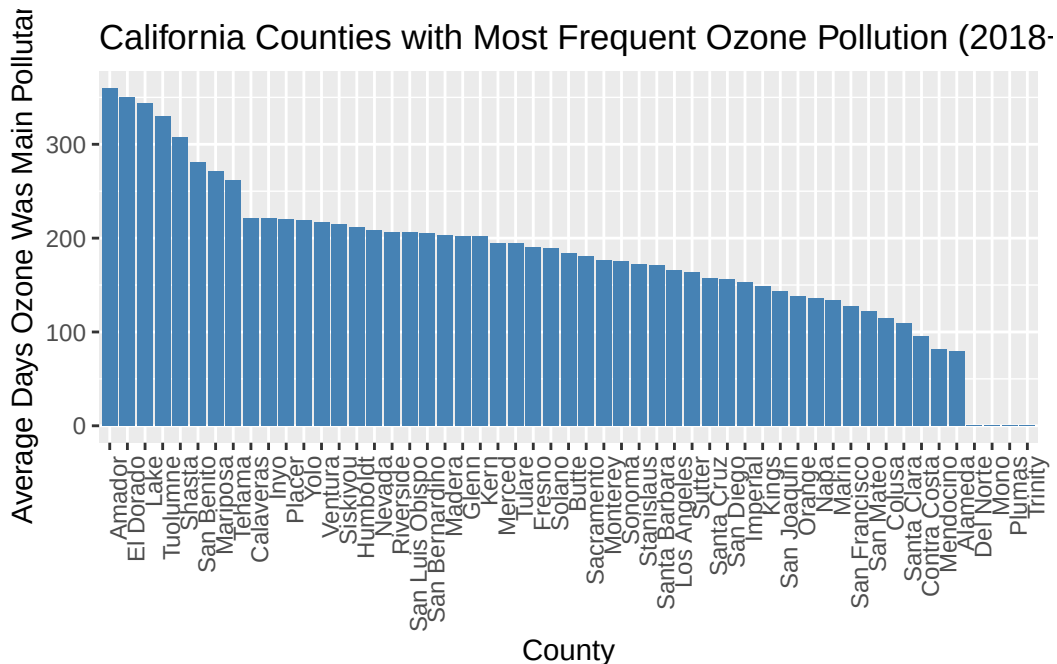
5 Shasta 307 2151 1435 2
6 San Benito 281 1969 174 1
7 Mariposa 271 1901 1073 8
8 Tehama 262 1839 218 6
9 Calaveras 221 1553 210 3
10 Inyo 221 1552 1551 7
# i 43 more rows
# i 3 more variables: avg_sensitive_group_days <int>,
# avg_very_unhealthy_days <int>, n_years_of_data <int>

```

```

ggplot(cali_county_ozone, aes(x = reorder(County, -avg_ozone_days), y = avg_ozone_days)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "California Counties with Most Frequent Ozone Pollution (2018-2024)",
       x = "County",
       y = "Average Days Ozone Was Main Pollutant") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

```



Now lets see which California counties that have the worst AQI(Air Quality Index).

```

cali_county_AQI <- Cali_all_years_data |>
  group_by(County) |>
  summarise(
    avg_median_aqi = mean(Median.AQI),

```

```

    avg_max_aqi = mean(Max.AQI),
    total_unhealthy_days = sum(Unhealthy.Days),
    total_sensitive_group_days = sum(Unhealthy.for.Sensitive.Groups.Days),
    avg_ozone_days = mean(Days.Ozone),
    n_years_of_data = n() # Number of years with data
  ) |>
  ungroup() |>
  arrange(desc(avg_median_aqi))

print(cali_county_AQI)

```

```

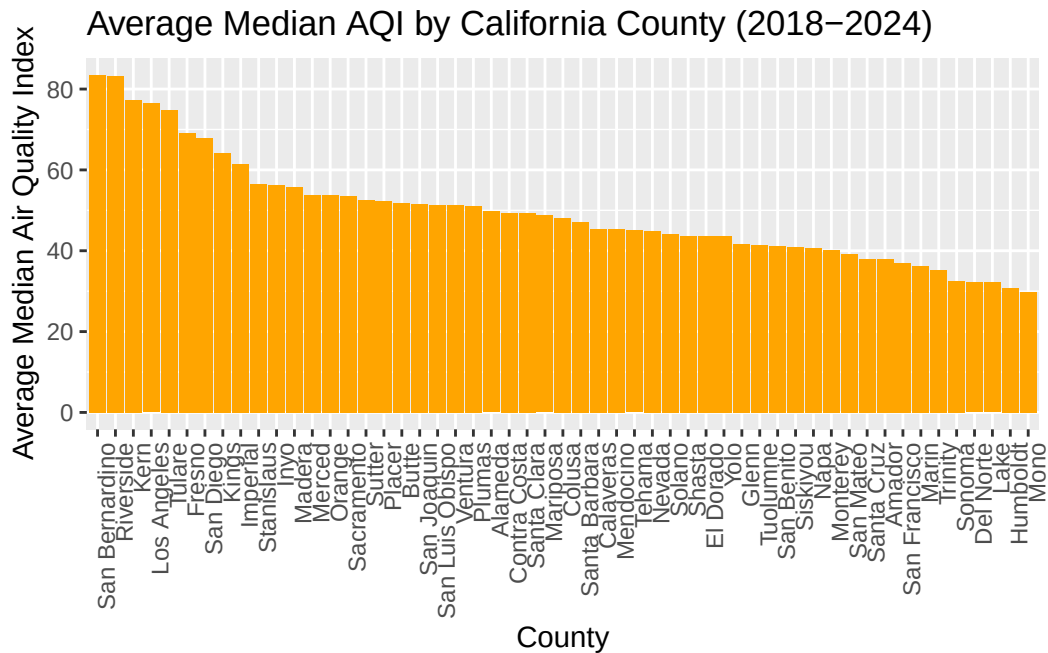
# A tibble: 53 x 7
  County avg_median_aqi avg_max_aqi total_unhealthy_days total_sensitive_grou~1
  <chr>      <dbl>      <dbl>          <int>          <int>
1 San B~      83.6      313.           396           506
2 River~      83.1      564.           265           619
3 Kern       77.3      281.           135           625
4 Los A~      76.4      226            221           488
5 Tulare      74.9      322.           151           619
6 Fresno      69.1      211.            89           452
7 San D~      67.9      158.            21           226
8 Kings       64.1      346            38           289
9 Imper~      61.4      387.            28           221
10 Stani~      56.6      182.            34           200
# i 43 more rows
# i abbreviated name: 1: total_sensitive_group_days
# i 2 more variables: avg_ozone_days <dbl>, n_years_of_data <int>

```

```

# Updated barchart
ggplot(cali_county_AQI, aes(x = reorder(County, -avg_median_aqi), y = avg_median_aqi)) +
  geom_bar(stat = "identity", fill = "orange") +
  labs(title = "Average Median AQI by California County (2018-2024)",
       x = "County",
       y = "Average Median Air Quality Index") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

```

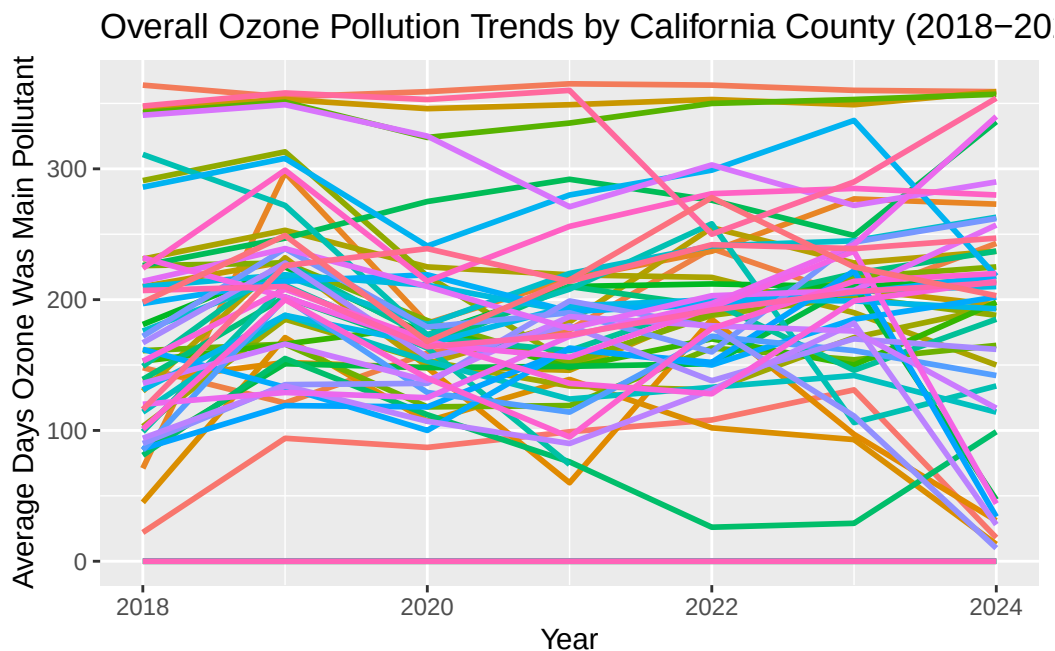
How has ozone changed over time?

```
ozone_trends <- Cali_all_years_data |>
  group_by(County, Year) |>
  summarise(ozone_days = mean(Days.Ozone)) |>
  ungroup() |>
  group_by(County) |>
  mutate(
    first_year = first(Year), # Gets the first year for each county
    last_year = last(Year),   # Gets the last year for each county
    first_ozone = first(ozone_days), # Ozone days in first year
    last_ozone = last(ozone_days)   # Ozone days in last year
  ) |>
  ungroup() |>
  distinct(County, first_ozone, last_ozone) |>
  mutate(change = last_ozone - first_ozone) |>
  arrange(desc(change))
```

`summarise()` has grouped output by 'County'. You can override using the `groups` argument.

```
# distinct() is used to ensure we have one row per county with the relevant data

# graph that represents overall ozone trends by county
ggplot(Cali_all_years_data, aes(x = Year, y = Days.Ozone, group = County, color = County)) +
  geom_line(linewidth = 1) +
  labs(title = "Overall Ozone Pollution Trends by California County (2018-2024)",
       x = "Year",
       y = "Average Days Ozone Was Main Pollutant") +
  theme(legend.position = "none")
```



```
# Counties that worsened the most (biggest positive change)
head(ozone_trends, 10)
```

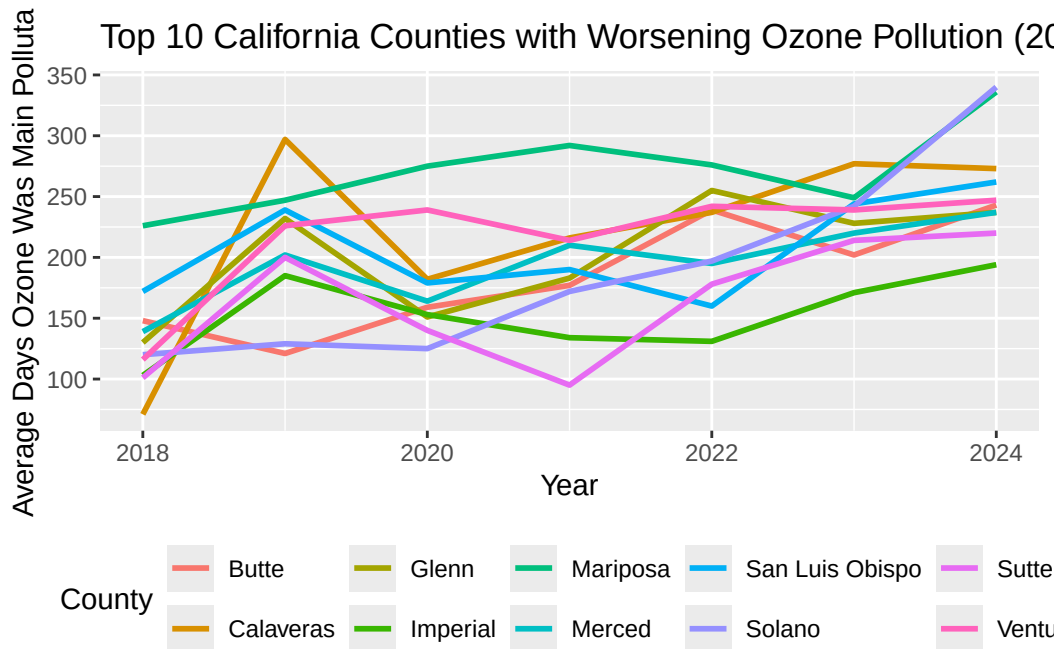
```
# A tibble: 10 x 4
  County      first_ozone last_ozone change
  <chr>      <dbl>      <dbl> <dbl>
1 Solano          120          340    220
2 Calaveras         71          273    202
3 Ventura         116          247    131
4 Sutter           101          220    119
5 Mariposa         226          336    110
6 Glenn           130          237    107
```

7 Merced	139	237	98
8 Butte	148	243	95
9 Imperial	103	194	91
10 San Luis Obispo	172	262	90

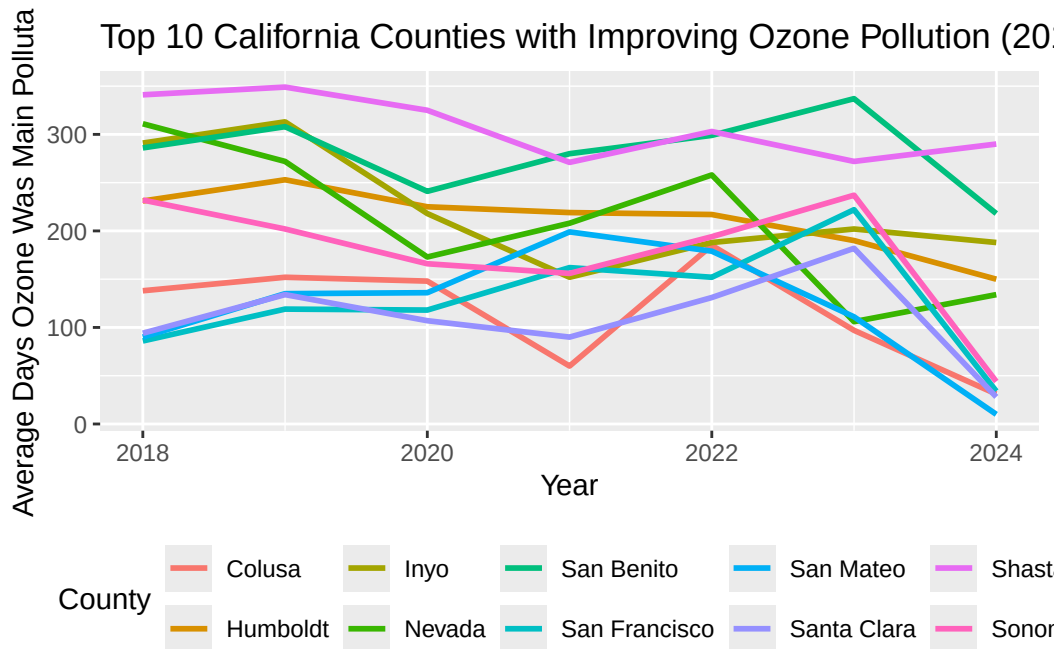
```
# Counties that improved the most (biggest negative change)
tail(ozone_trends, 10)
```

```
# A tibble: 10 x 4
  County      first_ozone last_ozone change
  <chr>      <dbl>      <dbl>  <dbl>
1 Shasta      341        290    -51
2 San Francisco    86         34   -52
3 Santa Clara     94         28  -66
4 San Benito    286        218  -68
5 San Mateo      90         10  -80
6 Humboldt    231        150  -81
7 Inyo        291        188 -103
8 Colusa      138         31 -107
9 Nevada     311        134 -177
10 Sonoma     232         44 -188
```

```
# graph that represents top 10 counties that worsened the most
ggplot(filter(Cali_all_years_data, County %in% head(ozone_trends$County, 10)),
  aes(x = Year, y = Days.Ozone, group = County, color = County)) +
  geom_line(linewidth = 1) +
  labs(title = "Top 10 California Counties with Worsening Ozone Pollution (2018-2024)",
    x = "Year",
    y = "Average Days Ozone Was Main Pollutant") +
  theme(legend.position = "bottom")
```



```
# graph that represents top 10 counties that improved the most
ggplot(filter(Cali_all_years_data, County %in% tail(ozone_trends$County, 10)),
  aes(x = Year, y = Days.Ozone, group = County, color = County)) +
  geom_line(linewidth = 1) +
  labs(title = "Top 10 California Counties with Improving Ozone Pollution (2018-2024)",
    x = "Year",
    y = "Average Days Ozone Was Main Pollutant") +
  theme(legend.position = "bottom")
```



Comparing Ozone vs Other Pollutants

```
# remember PM2.5 is fine particulate matter (2.5 micrometers)
# PM10 is coarse particulate matter (10 micrometers)
# CO is Carbon Monoxide
# NO2 is Nitrogen Dioxide
pollutant_comparison <- Cali_all_years_data |>
  group_by(County) |>
  summarise(
    avg_ozone = mean(Days.Ozone),
    avg_PM2.5 = mean(Days.PM2.5),
    avg_PM10 = mean(Days.PM10),
    avg_CO = mean(Days.CO),
    avg_NO2 = mean(Days.NO2),
  ) |>
  ungroup() |>
  arrange(desc(avg_ozone))

print(pollutant_comparison)
```

```
# A tibble: 53 x 6
  County      avg_ozone avg_PM2.5 avg_PM10 avg_CO avg_NO2
  <chr>      <dbl>    <dbl>    <dbl>  <dbl>  <dbl>
```

```

1 Amador      361.      0      0      0      0
2 El Dorado   351.     7.86   6.57   0      0
3 Lake        345.    13.3    1      0      0
4 Tuolumne    330.      0      0      0      0
5 Shasta      307.    57.4    0.429  0      0
6 San Benito  281.    80.6    3.43   0      0
7 Mariposa    272.    92.6    1.14   0      0
8 Tehama      263.   102     0.571  0      0
9 Calaveras   222.   140.    1.14   0      0
10 Inyo        222.   109    34.6    0      0
# i 43 more rows

```

Air Impact on Health

```

# Do counties with more ozone days have more unhealthy days?
health_analysis <- Cali_all_years_data |>
  group_by(County) |>
  summarise(
    avg_ozone_days = mean(Days.Ozone),
    avg_unhealthy_days = mean(Unhealthy.Days),
    avg_sensitive_days = mean(Unhealthy.for.Sensitive.Groups.Days),
    avg_very_unhealthy_days = mean(Very.Unhealthy.Days),
    avg_hazardous_days = mean(Hazardous.Days),
    total_unhealthy_days = sum(Unhealthy.Days + Unhealthy.for.Sensitive.Groups.Days +
                              Very.Unhealthy.Days + Hazardous.Days)
  ) |>
  ungroup() |>
  arrange(desc(avg_ozone_days))

head(health_analysis)

```

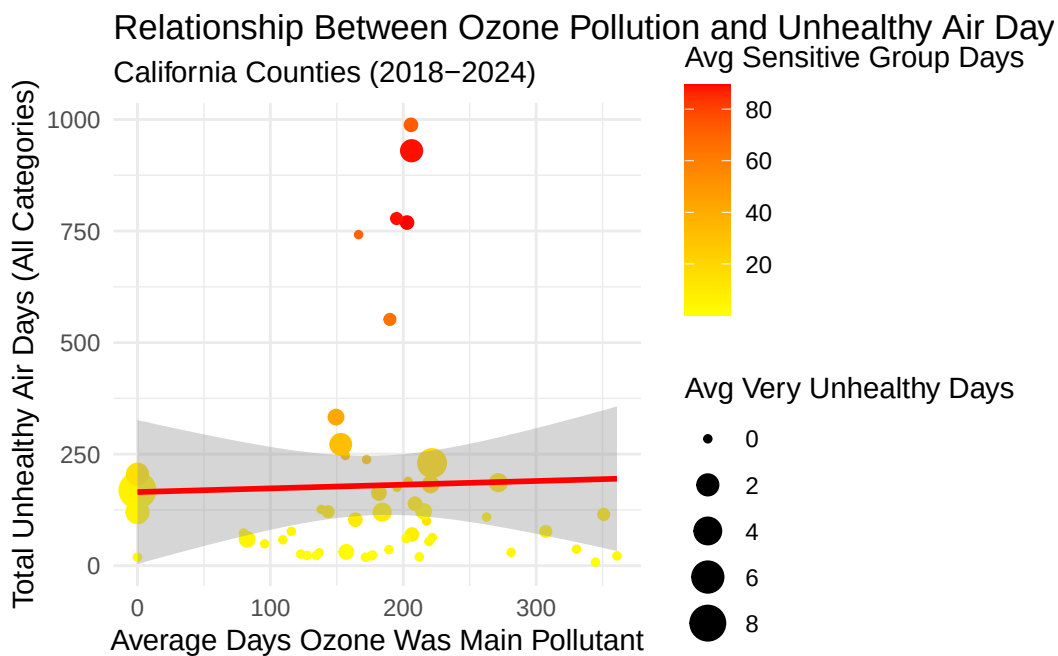
```

# A tibble: 6 x 7
  County      avg_ozone_days avg_unhealthy_days avg_sensitive_days
  <chr>          <dbl>          <dbl>          <dbl>
1 Amador      361.          0.143          3
2 El Dorado   351.          3             13
3 Lake        345.          0.714          0.286
4 Tuolumne    330.          0.143          5.14
5 Shasta      307.          2.71           7.14
6 San Benito  281.          1.29           3
# i 3 more variables: avg_very_unhealthy_days <dbl>, avg_hazardous_days <dbl>,
#   total_unhealthy_days <int>

```

```
# Scatter plot to visualize relationship
ggplot(health_analysis, aes(x = avg_ozone_days, y = total_unhealthy_days)) +
  geom_point(aes(size = avg_hazardous_days, color = avg_sensitive_days)) +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Relationship Between Ozone Pollution and Unhealthy Air Days",
    subtitle = "California Counties (2018-2024)",
    x = "Average Days Ozone Was Main Pollutant",
    y = "Total Unhealthy Air Days (All Categories)",
    size = "Avg Very Unhealthy Days",
    color = "Avg Sensitive Group Days"
  ) +
  theme_minimal() +
  scale_color_gradient(low = "yellow", high = "red")
```

`geom\_smooth()` using formula = 'y ~ x'



```
# categories based on ozone days
health_categories <- health_analysis |>
  mutate(ozone_level = case_when(
    avg_ozone_days > 300 ~ "Very High Ozone",
```

```

    avg_ozone_days > 200 ~ "High Ozone",
    avg_ozone_days > 100 ~ "Moderate Ozone",
    TRUE ~ "Low Ozone"
  ))

# health impacts across ozone levels
health_categories |>
  group_by(ozone_level) |>
  summarise(
    avg_bad_days = mean(total_unhealthy_days),
    county_count = n()
  ) |>
  ungroup()

```

```

# A tibble: 4 x 3
  ozone_level      avg_bad_days county_count
  <chr>          <dbl>         <int>
1 High Ozone      250.             17
2 Low Ozone       99.1             7
3 Moderate Ozone  181.            24
4 Very High Ozone  51.8             5

```

How did Covid-19 affect the ozone levels in California? pre-COVID (2018-2019), COVID (2020), and post-COVID (2021-2024) periods.

```

covid_analysis <- Cali_all_years_data |>
  mutate(
    time_period = case_when(
      Year <= 2019 ~ "Pre-COVID (2018-2019)",
      Year == 2020 ~ "COVID Lockdown (2020)",
      Year >= 2021 ~ "Post-COVID (2021-2024)"
    )
  )

covid_summary <- covid_analysis |>
  group_by(time_period) |>
  summarise(
    avg_ozone_days = mean(Days.Ozone),
    avg_median_aqi = mean(Median.AQI),
    avg_max_aqi = mean(Max.AQI),
    avg_unhealthy_days = mean(Unhealthy.Days)
  )

```



```

) |>
ungroup() |>
  arrange(factor(time_period, levels = c("Pre-COVID (2018-2019)",
                                         "COVID Lockdown (2020)",
                                         "Post-COVID (2021-2024)")))
# to arrange from 2019 to 2024, we had to ungroup() first and then use
# factor() inside arrange(), what factor() does is that it allows us to
# specify the order of the levels

print(covid_summary)

```

```

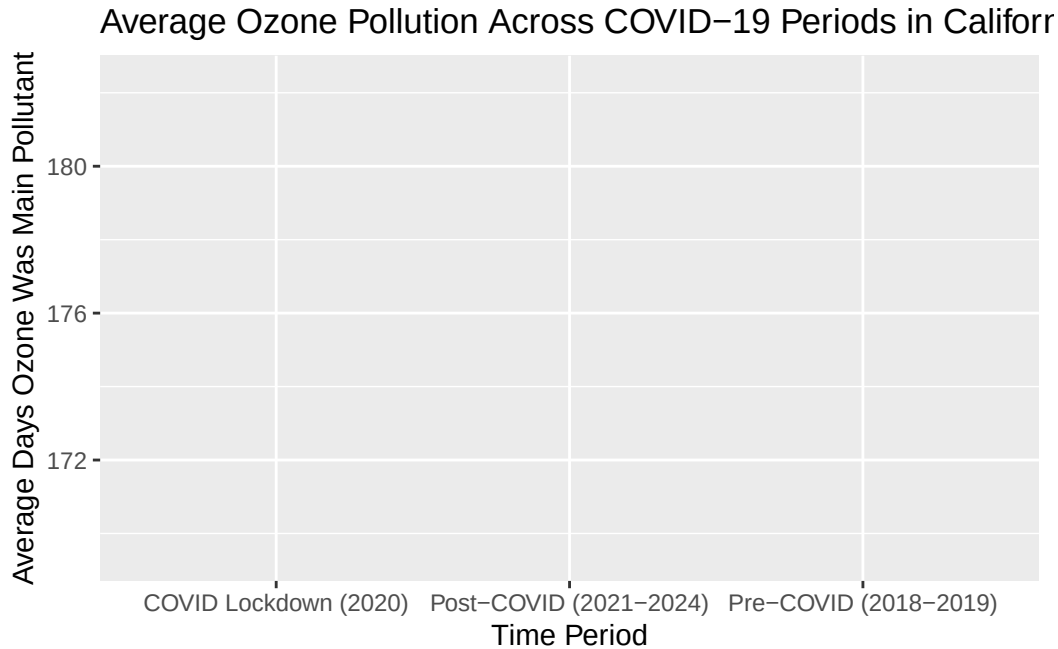
# A tibble: 3 x 5
  time_period      avg_ozone_days avg_median_aqi avg_max_aqi avg_unhealthy_days
  <chr>              <dbl>         <dbl>         <dbl>         <dbl>
1 Pre-COVID (2018-- 182.          49.6          243.           5.86
2 COVID Lockdown (~ 169.          51.8          331.          13.2
3 Post-COVID (2021~ 182.          49.0          269.           5.10

```

```

ggplot(covid_summary, aes(x = time_period, y = avg_ozone_days,
                          fill = time_period)) +
  labs(
    title = "Average Ozone Pollution Across COVID-19 Periods in California",
    x = "Time Period",
    y = "Average Days Ozone Was Main Pollutant"
  ) +
  theme(legend.position = "none")

```



```
# here "fill = time_period" is used to color the bars differently based on
# the time period
```

comparing Urban and Rural Counties (to compare ozone levels

based on traffic density)

```
urban_counties <- c("Los Angeles", "San Francisco", "San Diego",
  "Santa Clara", "Alameda", "Orange")
rural_counties <- c("Alpine", "Amador", "Calaveras", "Glenn",
  "Humboldt", "Inyo", "Lake", "Mariposa", "Mendocino",
  "Modoc", "Mono", "Nevada", "Plumas", "Sierra",
  "Siskiyou", "Tehama", "Trinity", "Tuolumne")

covid_urban_rural <- covid_analysis |>
  mutate(
    county_type = case_when(
      County %in% urban_counties ~ "Urban",
      County %in% rural_counties ~ "Rural",
      TRUE ~ "Mixed")
  ) |>
```

```
group_by(time_period, county_type) |>
summarise(
  avg_ozone = mean(Days.Ozone),
  avg_aqi = mean(Median.AQI),
  n_counties = n_distinct(County)
) |>
ungroup() |>
arrange(county_type)
```

`summarise()` has grouped output by 'time_period'. You can override using the `.groups` argument.

```
print(covid_urban_rural)
```

```
# A tibble: 9 x 5
  time_period      county_type avg_ozone avg_aqi n_counties
  <chr>           <chr>      <dbl>   <dbl>    <int>
1 COVID Lockdown (2020) Mixed        170.    54.2      32
2 Post-COVID (2021-2024) Mixed        186.    51.6      32
3 Pre-COVID (2018-2019) Mixed        182.    52.3      32
4 COVID Lockdown (2020) Rural         186.    43.5      15
5 Post-COVID (2021-2024) Rural         192.    41.6      15
6 Pre-COVID (2018-2019) Rural         207.    41.2      15
7 COVID Lockdown (2020) Urban         124.    60.2       6
8 Post-COVID (2021-2024) Urban         135.    54.0       6
9 Pre-COVID (2018-2019) Urban         122.    56.8       6
```

```
# now lets see specific counties with the biggest changes during lockdown
covid_changes <- covid_analysis |>
group_by(County, time_period) |>
summarise(avg_ozone = mean(Days.Ozone)) |>
pivot_wider(names_from = time_period, values_from = avg_ozone) |>
mutate(
  net_change = `Post-COVID (2021-2024)` - `Pre-COVID (2018-2019)`,
  lockdown_change = `COVID Lockdown (2020)` - `Pre-COVID (2018-2019)`,
  post_lockdown_change = `Post-COVID (2021-2024)` - `COVID Lockdown (2020)` |>
ungroup() |>
relocate(net_change, lockdown_change, post_lockdown_change,
  .before = `COVID Lockdown (2020)`) |>
arrange(desc(abs(net_change)))
```

`summarise()` has grouped output by 'County'. You can override using the
 `.groups` argument.

```
# we use `` in mutate here because the column names have spaces and
# special characters, otherwise we don't need it
# if the net change is positive, that means ozone levels increased
# from pre-COVID to post-COVID
# used new relocate function to move the "Change" columns to the front,
# I put .before `COVID Lockdown (2020)` because I knew it was after County,
# and it was quicker than writing every other column name
```

```
print(head(covid_changes, 20))
```

```
# A tibble: 20 x 7
```

	County	net_change	lockdown_change	post_lockdown_change	COVID Lockdown (2020~1
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Inyo	-120.	-84	-35.5	218
2	Nevada	-115	-118.	3.5	173
3	Solano	113.	0.5	113.	125
4	Napa	-81	8	-89	163
5	Butte	80.8	24.5	56.2	159
6	Calav~	66.8	-2	68.8	182
7	Ventu~	64.5	68	-3.5	239
8	Shasta	-61	-20	-41	325
9	Mendo~	-60.5	-6	-54.5	112
10	Sonoma	-59.2	-51	-8.25	166
11	Colusa	-51.8	3	-54.8	148
12	Marip~	51.8	38.5	13.2	275
13	Humbo~	-48	-17	-31	225
14	Placer	45.2	-16	61.2	181
15	Merced	45	-6.5	51.5	164
16	Glenn	44.8	-30	74.8	151
17	San F~	40	15.5	24.5	118
18	Tuolu~	-39.5	0	-39.5	353
19	Sacra~	37.2	6.5	30.8	166
20	Santa~	-35	-41.5	6.5	156

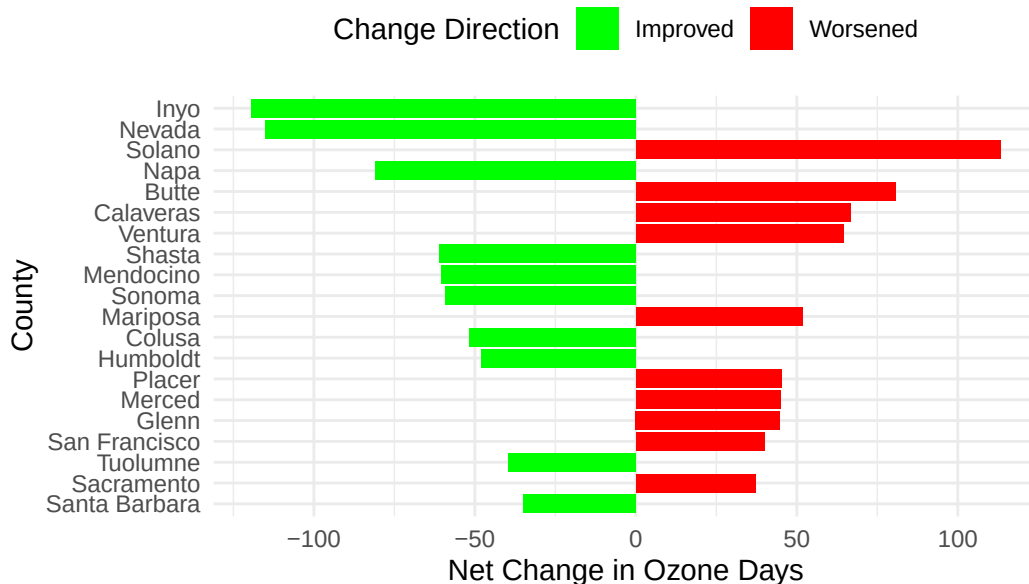
```
# i abbreviated name: 1: `COVID Lockdown (2020)`
```

```
# i 2 more variables: `Post-COVID (2021-2024)` <dbl>,
```

```
# `Pre-COVID (2018-2019)` <dbl>
```

```
# this barplot shows the top 20 counties with the biggest net changes
ggplot(head(covid_changes, 20), aes(x = reorder(County, abs(net_change)),
                                     y = net_change)) +
  geom_bar(stat = "identity", aes(fill = net_change > 0)) +
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "green"),
                    labels = c("Improved", "Worsened"),
                    name = "Change Direction") +
  labs(
    title = "Top 20 Changes in Ozone Days: Pre-COVID vs Post-COVID",
    x = "County",
    y = "Net Change in Ozone Days"
  ) +
  coord_flip() +
  theme_minimal() +
  theme(legend.position = "top")
```

Top 20 Changes in Ozone Days: Pre-COVID vs Post-COVID



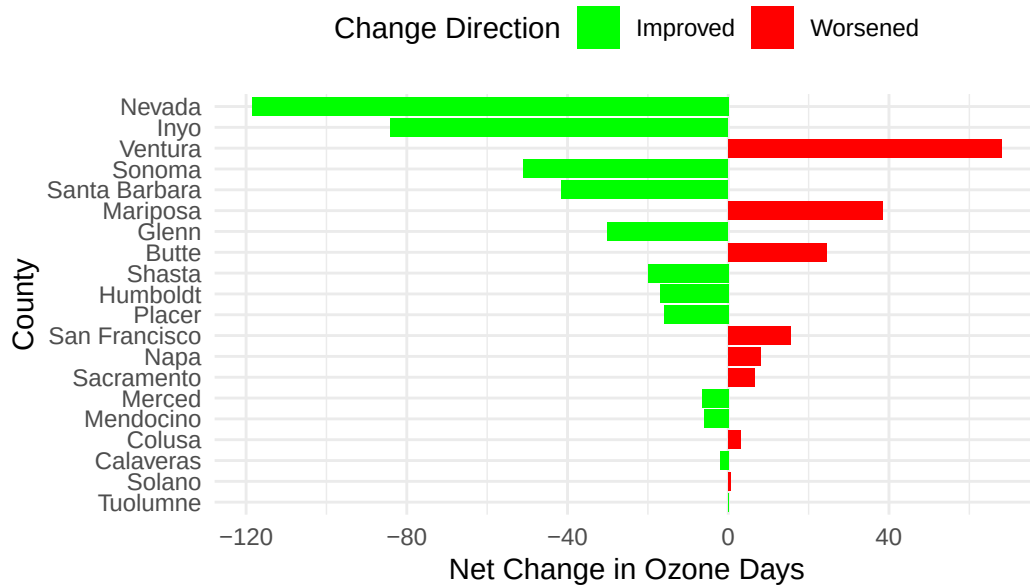
```
# this barplot shows the top 20 lockdown biggest changes (2018-2020)
ggplot(head(covid_changes, 20), aes(x = reorder(County, abs(lockdown_change)),
                                     y = lockdown_change)) +
  geom_bar(stat = "identity", aes(fill = lockdown_change > 0)) +
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "green"),
                    labels = c("Improved", "Worsened"),
```

```

    name = "Change Direction") +
  labs(
    title = "Top 20 Changes in Ozone Days: Pre-COVID vs COVID-Lockdown",
    x = "County",
    y = "Net Change in Ozone Days"
  ) +
  coord_flip() +
  theme_minimal() +
  theme(legend.position = "top")

```

Top 20 Changes in Ozone Days: Pre-COVID vs COVID



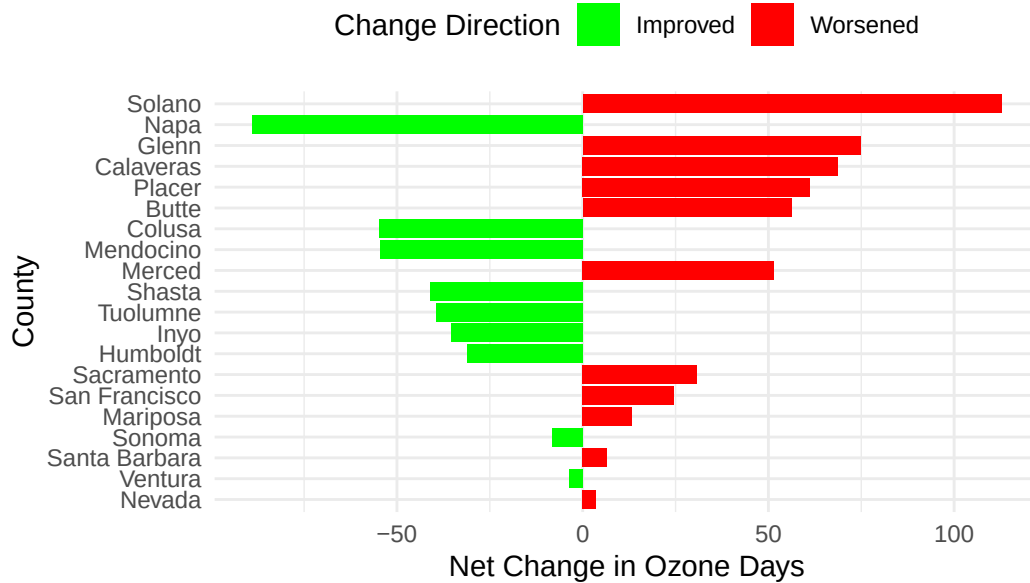
```

# this barplot shows the top 20 post-lockdown biggest changes (2020-2024)
ggplot(head(covid_changes, 20), aes(x = reorder(County,
    abs(post_lockdown_change)),
    y = post_lockdown_change)) +
  geom_bar(stat = "identity", aes(fill = post_lockdown_change > 0)) +
  scale_fill_manual(values = c("TRUE" = "red", "FALSE" = "green"),
    labels = c("Improved", "Worsened"),
    name = "Change Direction") +
  labs(
    title = "Top 20 Changes in Ozone Days: COVID-Lockdown vs Post-Covid",
    x = "County",
    y = "Net Change in Ozone Days"
  )

```

```
) +
coord_flip() +
theme_minimal() +
theme(legend.position = "top")
```

Top 20 Changes in Ozone Days: COVID–Lockdown vs



Statistical Test to see if the Covid lockdown made a significant difference

```
# Did the lockdown really make a difference?
pre_covid <- covid_analysis %>% filter(time_period == "Pre-COVID (2018-2019)") %>% pull(Days)
during_covid <- covid_analysis %>% filter(time_period == "COVID Lockdown (2020)") %>% pull(Days)

t_test_result <- t.test(pre_covid, during_covid)
print(t_test_result)
```

Welch Two Sample t-test

```
data: pre_covid and during_covid
t = 0.91216, df = 118.32, p-value = 0.3635
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-15.22189 41.22189
```

```
sample estimates:
mean of x mean of y
 182.3585  169.3585
```

```
# p-value = 0.3635 → Not statistically significant
# Mean Pre-COVID: 182.4 ozone days
# Mean During COVID: 169.4 ozone days
# Difference: 13 days (7% decrease)
# While there was a decrease during lockdowns, the variation between
# counties was so large that we can't confidently say this wasn't just
# random fluctuation.
```

Statistical Test to see if the Covid lockdown made a significant difference (rural vs urban)

```
# Calculate how many counties improved vs worsened during COVID
covid_summary_stats <- covid_changes |>
  summarise(
    counties_improved = sum(net_change < 0),
    counties_worsened = sum(net_change > 0),
    avg_net_change = mean(net_change),
    max_improvement = min(net_change),
    max_worsening = max(net_change)
  )

print(covid_summary_stats)
```

```
# A tibble: 1 x 5
  counties_improved counties_worsened avg_net_change max_improvement
      <int>           <int>           <dbl>           <dbl>
1           23           26           -1.75           -120.
# i 1 more variable: max_worsening <dbl>
```